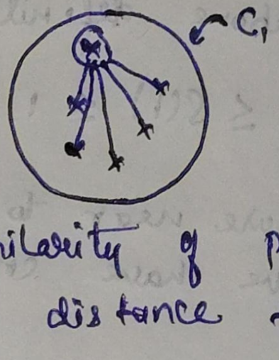
Silhoutte (clustering)

i. For ~~day~~ data point $i \in C_1$ (data point i in the cluster C_1), let

$$a(i) = \frac{1}{|C_1| - 1} \sum_{j \in C_2, i \neq j} d(i, j).$$

be the mean distance between i and all other data points in the same cluster, where $|C_1|$ is the number of points belonging to cluster 1, and $d(i, j)$ is the distance between data points i and j in the cluster C_2 (we divide by $|C_2| - 1$ because we do not include the distance $d(i, j)$ in the sum). We can interpret $a(i)$ as a measure of how well i is assigned to its cluster (the smaller the value, the better the assignment).

$$\boxed{a(i) = \frac{1}{|C_1| - 1} \sum_{j \in C_2, i \neq j} d(i, j)}$$

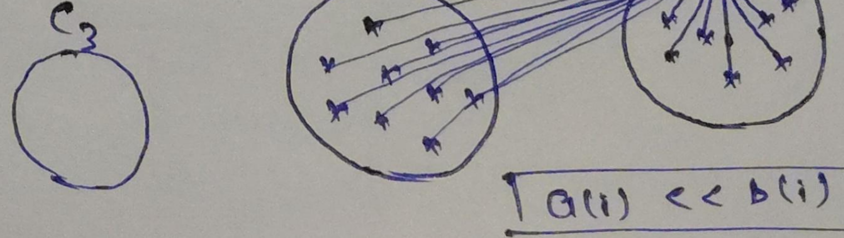


ii. We then define the mean dissimilarity of point i to some cluster C_j as the mean of the distance from i to all points in C_j (where $C_j \neq C_i$).

For each data point $i \in C_i$, we now define

$$\boxed{b(i) = \min_{j \neq i} \frac{1}{|C_j|} \sum_{j \in C_j} d(i, j)}$$

to be the smallest (hence the min operator in the formula) mean distance of i to all points in any other cluster, of which i is not a member. The cluster with this smallest mean dissimilarity is said to be the "neighboring cluster" of i because it is the next best fit cluster ~~of~~ for point i



$$\boxed{a(i) << b(i)}$$

III. Silhouette Score

we now define a silhouette (value) of one data point i

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \quad \text{if } |C_i| > 1$$

and

$$S(i) = 0, \quad \text{if } |C_i| = 1$$

which can be also written as:

$$S(i) = \begin{cases} 1 - \underbrace{a(i)/b(i)}_{\approx 1} & \text{, if } a(i) < b(i) \\ 0 & \text{if } a(i) = b(i) \\ \underbrace{b(i)/a(i) - 1}_{\approx -1} & \text{, if } a(i) > b(i) \end{cases}$$

From the above definition it is clear that

$$-1 \le S(i) \le 1$$

$\underline{[-1 \text{ to } 1]}$ more near to 1 better clustering model we have created.